Neel M Naik

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PROFILE OVERVIEW

Results-driven Information Science Engineering student specializing in machine learning, 3D computer vision, and algorithmic optimization. Proven expertise in developing PyTorch models for LiDAR point-cloud processing, alongside engineering high-contention scheduling engines and secure backend utilities. Adept at transforming complex spatial datasets and strict constraints into scalable, interactive software solutions.

EDUCATION

Nitte Meenakshi Institute of Technology, Bengaluru *2021 – Present*
B.E. in Information Science and Engineering — [CGPA / Percentage]

Narayana PU College, Bengaluru *2017 – 2019*
Class XII, State Board, 43.67%

WORK EXPERIENCE

Intern — Bharat Electronics Limited (BEL), Bengaluru *Oct 2025 – Nov 2025*
Product Development and Innovation Centre (PDIC)

- *Worked on a project titled "Secure File Encryption Utility" — addressed the critical risk of exposed local data by developing a Python encryption tool that leverages the Advanced Encryption Standard to scramble plaintext into unreadable ciphertext.*
- *Developed a standalone file security tool using Python and the cryptography package to execute AES-128 symmetric encryption, secure key storage, and lossless decryption for localized plaintext files.*
- *Successfully validated lossless data restoration while mastering symmetric key lifecycles, the CIA triad, and defense-standard data handling protocols.*

Machine Learning Engineer, Intern — Bellatrix Aerospace, Bengaluru *Mar 2026 – Jun 2026*

- *Engineered a 3D point-cloud registration pipeline using PointNet autoencoders, PCA, and ICP algorithms to identify and estimate satellite poses from LiDAR spatial datasets.*
- *Processed and trained models on datasets containing over **20,000+** spatial data points, reducing rotational pose estimation errors to **< 4 degrees** to support autonomous rendezvous simulations.*

PROJECTS CREATED

Interactive 3D Virtual Environment *Jul 2024*
C#, Unity Engine, 3D Spatial Design

- *Developed an immersive, interactive 3D virtual environment from scratch using the Unity engine and object-oriented C# scripting.*
- *Programmed custom user controls, player kinematics, and real-time environmental physics to create a seamless and responsive user experience.*
- *Designed and optimized spatial layouts and rendering pipelines to maintain high frame rates and smooth interactions within the simulated world.*

Secure File Encryption Utility *Nov 2025*
Python 3.x, Visual Studio Code, Git

- *Engineered a standalone file encryption utility in Python using the cryptography library (Fernet/AES-128) to secure local plaintext data at rest against unauthorized physical access.*
- *Implemented automated 128-bit symmetric key generation, safe key storage, and binary file I/O operations for seamless file overwriting.*
- *Validated end-to-end encryption and decryption pipelines, ensuring 100% data confidentiality and zero data corruption during data restoration.*

Satellite Identification via 3D Point Clouds *Jun 2026*
Python, PyTorch, PointNet, PCA, ICP, Google Colab

- *Engineered an end-to-end machine learning pipeline utilizing PointNet autoencoders to accurately classify and register 3D satellite models from raw LiDAR spatial datasets.*
- *Implemented Principal Component Analysis (PCA) and Iterative Closest Point (ICP) algorithms from scratch to optimize point-cloud alignment and 3D reconstruction accuracy.*
- *Processed high-dimensional spatial data within Google Colab, successfully reducing point-cloud registration errors and ensuring high-fidelity rendering for aerospace simulations.*

Dynamic Placement Week Scheduler & Replanning Engine *Aug 2026*
Python 3.11, Streamlit, Pandas, Heuristic Optimization, Constraint Satisfaction Problems (CSP), OOP

- *Engineered a discrete-event heuristic scheduling engine coordinating high-contention campus recruitment drives across **35 companies, 800 candidates, and 20 interview rooms** over a 4-day period.*
- *Implemented priority-driven slot allocation algorithms enforcing strict hard constraints, achieving **zero double-booking** across candidates, interview rooms, and multi-member evaluation panels.*
- *Developed a real-time replanning mechanism to dynamically handle operational disruptions (delayed recruiters, panel dropouts, room outages, and candidate withdrawals) while bounding schedule reshuffling strictly to affected entities.*
- *Built and deployed an interactive Streamlit control room featuring automated constraint-verification test suites, live conflict detection, and structured diff logging, ensuring **100% mathematical clash prevention**.*

SKILLS

Technical Skills: Python, PyTorch, C#, C, SQL, Deep Learning (PointNet, CNNs, Autoencoders), 3D Computer Vision (ICP, PCA, LiDAR Processing), Cryptography (AES-256), Heuristic Optimization, Constraint Satisfaction Problems (CSP), Object-Oriented Programming (OOP), Data Structures & Algorithms

Tools & Platforms: Git, GitHub, Google Colab, VS Code, Unity Engine, Streamlit, Linux/Unix, LaTeX, AWS (Cloud Fundamentals)

Soft Skills: Complex Problem-Solving, Analytical Thinking, Cross-Functional Collaboration, Technical Documentation & Reporting, Agile Adaptability, System Design & Planning

ACTIVITIES

Strength Training *2018 – Present*

- *Dedicated to a multi-year fitness and strength training regimen.*